

T1.EM3 Circuits

Vijay Hans, Prerit Bucha

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1 Introduction

This report presents an experimental investigation of transient voltage behavior in RC and RLC circuits across 4 experiments.

In Experiment 1A, we measured the time constant τ of an RC circuit with $R = 10 \text{ k}\Omega$ and $C = 47 \text{ }\mu\text{F}$ driven by an AC voltage source. To accomplish this, we measured the voltage across the capacitor with an oscilloscope and compared it to the theoretically calculated value.

In Experiment 1B, we fashioned a homemade capacitor out of a sheet of paper and two strips of aluminum foil. We calculated a theoretical capacitance based on an assumed dielectric constant for paper and measurements of the dimensions of the capacitor. We then compared this with an experimental observation where we placed the capacitor in series with a $1 \text{ M}\Omega$ resistor and an AC voltage source to measure the time constant and from that extract the capacitance.

In Experiment 2A, we considered an underdamped RLC circuit and the emergent transient responses. We connected in series a 100 mH inductor, $0.1 \text{ }\mu\text{F}$ capacitor, and two parallelly-connected $100 \text{ }\Omega$ resistors to an AC voltage source and fit our observations to expectations.

In Experiment 2B, we considered an overdamped RLC circuit and the emergent transient responses. We connected in series a 100 mH inductor, $47 \text{ }\mu\text{F}$ capacitor, and a $10 \text{ k}\Omega$ resistor to an AC voltage source and fit our observations to expectations.

2 Theory

RC Circuits with a DC Source

The RC circuit studied here consists of a resistor R and capacitor C connected in series with a DC source V_0 . A switch toggles between two positions: Position 1 charges the capacitor through the resistor; Position 2 allows the capacitor to discharge through the resistor.

Charging Capacitor (Switch Position 1)

With the source in the loop, Kirchhoff's voltage law gives:

$$V_0 - V_R - V_C = 0 \tag{2.0.1}$$

which leads to the ODE:

$$\frac{dQ}{dt} = -\frac{1}{RC} Q + \frac{V_0}{R} \quad (2.0.2)$$

With initial condition $Q(0) = 0$, the capacitor voltage is:

$$V_C(t) = V_0 \left(1 - e^{-t/\tau}\right) \quad (2.0.3)$$

where the **time constant** is defined as:

$$\tau \equiv RC \quad (2.0.4)$$

As $t \rightarrow \infty$, $Q(t) \rightarrow CV_0$ and $I(t) \rightarrow 0$.

For the purposes of regression, we may rewrite this equation as:

$$\frac{V_C(t)}{V_0} = A(1 - e^{-t/B}) \quad (2.0.5)$$

where A is a constant describing amplitude that we expect to be 1, and B is a constant describing decay time that we expect to be τ .

Discharging Capacitor (Switch Position 2)

Applying Kirchhoff's voltage law around the discharge loop gives:

$$-V_R - V_C = 0 \quad (2.0.6)$$

Substituting Ohm's law $V_R = IR$, the capacitor relation $V_C = Q/C$, and the current definition $I = dQ/dt$ yields a first-order linear ODE:

$$\frac{dQ}{dt} = -\frac{1}{RC} Q \quad (2.0.7)$$

With the initial condition $Q(0) = Q_0 = CV_0$, the solution is:

$$Q(t) = CV_0 e^{-t/\tau} \quad (2.0.8)$$

and the capacitor voltage is:

$$V_C(t) = V_0 e^{-t/\tau} \quad (2.0.9)$$

where $\tau = RC$ as before. As $t \rightarrow \infty$, both $Q(t) \rightarrow 0$ and $I(t) \rightarrow 0$.

As before, the regression model is:

$$\frac{V_C(t)}{V_0} = A e^{-t/B} \quad (2.0.10)$$

where, as before, we expect A to be 1 and B to be τ .

Capacitors, Capacitance, and Breakdown Voltage

A **capacitor** is a two-terminal device consisting of two conductors separated by a dielectric layer that stores electrical charge and energy. When the conductors carry a charge difference Q , a voltage difference V_C appears between them. The **capacitance** is defined as the ratio of charge to voltage:

$$C = \frac{Q}{V_C} \quad (2.0.11)$$

The simplest capacitor is the **parallel-plate capacitor**, consisting of two plates of equal area A separated by a distance d of dielectric material with relative static permittivity ε_r . Its capacitance is:

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d} \quad (2.0.12)$$

where $\varepsilon_0 = 8.854 \times 10^{-12} \text{ F m}^{-1}$ is the permittivity of free space.

If the electric field inside the dielectric exceeds a breakdown threshold E_{bd} , **dielectric breakdown** occurs and charge flows through the dielectric. Real capacitors are rated by their **breakdown voltage** $V_{bd} = E_{bd} d$; applying a voltage greater than V_{bd} may permanently damage the capacitor.

Series RLC Circuits

When an inductor L is added in series with R and C , the circuit behaves as an electronic analogue of a damped harmonic oscillator. Energy oscillates between the electric field of the capacitor and the magnetic field of the inductor, while being dissipated by the resistor.

Transient behavior is the response observed when the circuit input undergoes a short-lived step change and is then held constant.

Applying Kirchhoff's voltage law to the series RLC loop with source voltage V_0 :

$$V_0 - V_R - V_L - V_C = 0 \quad (2.0.13)$$

Using $V_R = IR$, $V_L = L dI/dt$, and $V_C = Q/C$, one obtains the second-order ODE for charge:

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = V_0 \quad (2.0.14)$$

Differentiating with respect to time gives a homogeneous equation for the current $I = dQ/dt$:

$$L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = 0 \quad (2.0.15)$$

We define two characteristic frequencies:

$$\omega_0 \equiv \frac{1}{\sqrt{LC}}, \quad \alpha \equiv \frac{R}{2L} \quad (2.0.16)$$

The behavior of equation (2.0.15) depends on the ratio α/ω_0 :

- **Underdamped** ($\alpha < \omega_0$): the circuit oscillates at angular frequency

$$\omega = \sqrt{\omega_0^2 - \alpha^2} \quad (2.0.17)$$

with exponentially decaying amplitude. The current is:

$$I(t) = e^{-\alpha t}(A \cos \omega t + B \sin \omega t) = \tilde{A} e^{-\alpha t} \cos(\omega t - \varphi) \quad (2.0.18)$$

- **Critically damped** ($\alpha = \omega_0$): the fastest non-oscillatory decay:

$$I(t) = e^{-\alpha t}(A + Bt) \quad (2.0.19)$$

- **Overdamped** ($\alpha > \omega_0$): purely exponential decay with no oscillations, governed by the parameter

$$b = \sqrt{\alpha^2 - \omega_0^2} \quad (2.0.20)$$

The current is:

$$I(t) = e^{-\alpha t}(Ae^{+bt} + Be^{-bt}) = A' e^{-(\alpha-b)t} + B' e^{-(\alpha+b)t} \quad (2.0.21)$$

where $A' = Ae^0$ and $B' = Be^0$ are determined by initial conditions. The **slow mode** decays with rate $D_{\text{slow}} = \alpha - b$ (time constant $\tau_{\text{slow}} = 1/(\alpha - b)$) and the **fast mode** decays with rate $D_{\text{fast}} = \alpha + b \gg D_{\text{slow}}$.

Decay Envelopes in the Underdamped Regime

In the underdamped case, the current oscillates while its amplitude decays exponentially. The instantaneous peaks of $I(t)$ lie on a smooth decay envelope obtained by noting that the cosine factor in equation (2.0.18) equals ± 1 at each successive extremum. At those moments the full solution reduces to

$$|I(t_n)| = \tilde{A} e^{-\alpha t_n}, \quad (2.0.22)$$

where t_n is the time of the n -th peak and $\tilde{A}$ is the overall oscillation amplitude set by the initial conditions. The envelope is therefore a pure exponential with decay constant $\alpha = R/2L$, independent of ω_0 or ω .

This separation is practically useful: fitting the envelope to the measured peak amplitudes isolates α without requiring knowledge of the full waveform, and the oscillation frequency ω can be recovered independently from the average peak-to-peak spacing $T = 2\pi/\omega$. Together, α and ω determine ω_0 via

$$\omega_0 = \sqrt{\omega^2 + \alpha^2}. \quad (2.0.23)$$

Error Propagation

The theoretical time constant $\tau = RC$ carries uncertainty from the component tolerances. Using standard propagation of independent uncertainties:

$$\frac{\delta\tau}{\tau} = \sqrt{\left(\frac{\delta R}{R}\right)^2 + \left(\frac{\delta C}{C}\right)^2} \quad (2.0.24)$$

For the RLC parameters defined in equation (2.0.16):

$$\frac{\delta\omega_0}{\omega_0} = \frac{1}{2} \sqrt{\left(\frac{\delta L}{L}\right)^2 + \left(\frac{\delta C}{C}\right)^2}, \quad \frac{\delta\alpha}{\alpha} = \sqrt{\left(\frac{\delta R}{R}\right)^2 + \left(\frac{\delta L}{L}\right)^2} \quad (2.0.25)$$

Component tolerances used in this lab are: resistors $\pm 1\%$, capacitors $\pm 5\%$, and inductor $\pm 10\%$.

Experimental values are extracted by fitting the models in equations (2.0.9)–(2.0.21) to oscilloscope data using nonlinear least squares. Agreement between an experimental value $x_{\text{exp}} \pm \delta x_{\text{exp}}$ and a theoretical value $x_{\text{th}} \pm \delta x_{\text{th}}$ is confirmed if the following holds:

$$|x_{\text{exp}} - x_{\text{th}}| < 2\sqrt{(\delta x_{\text{exp}})^2 + (\delta x_{\text{th}})^2} \quad (2.0.26)$$

3 Experiment 1A: Measuring the RC Time Constant with $R = 10\text{ k}\Omega$, $C = 47\text{ }\mu\text{F}$

3.1 Setup

Our primary objective is to measure the time constant of an RC circuit with specified circuit components. The setup for the lab is outlined as follows:

3.1.1 Materials

For this section of the lab, we used:

- an MSO 2024B Oscilloscope,
- a Tektronix AFG2021 Function Generator,
- a breadboard,
- a $10\text{ k}\Omega$ resistor with a $\pm 1\%$ tolerance,
- a $47\text{ }\mu\text{F}$ capacitor with a $\pm 5\%$ tolerance,
- and various leads, grabbers, and BNC cables.

3.1.2 Function Generator

The objective was to create a low-frequency square wave to serve as our power source. To do so, we:

- chose “Square Wave” as our desired signal output,
- set the frequency to 100 mHz ,
- set the output load impedance to Hi-Z,
- set the amplitude to 10 Vpp ,
- set the offset to 5 V (this ensures that the voltage generated is always nonnegative),
- and connected a BNC T-splitter to the leftmost generator output, intending to feed one end into the oscilloscope and use the other end as the power source in our breadboard circuit.

3.1.3 Oscilloscope

Here, we hoped to measure both the function generator output and the voltage difference across the capacitor. To accomplish this, we:

- displayed channels 1 and 2 on the scope,
- connected one BNC cable from the T-splitter to the oscilloscope CH1 input,
- connected a mini-grabber to the remaining BNC cable from the other end of the T-splitter (intending to connect to our circuit as an AC source),
- and connected another BNC cable to CH2, as well as to a mini-grabber (intending to connect across the capacitor). It is relevant to note that, since the signal has a frequency of only 100 mHz, the oscilloscope cannot trigger repetitively in auto mode. We instead used roll mode and pressed the “Run/Stop” button to freeze the screen on points of interest.

3.1.4 Breadboard/RC Circuit

Here, we hoped to create the circuit shown below on the breadboard. The voltmeters represent connections to the oscilloscope.

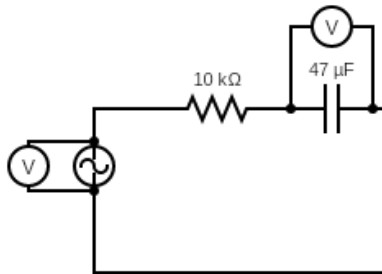


Figure 3.1.1: Circuit diagram for the setup in this experiment.

We used the circuit elements outlined in 3.1.1 to construct the following circuit:

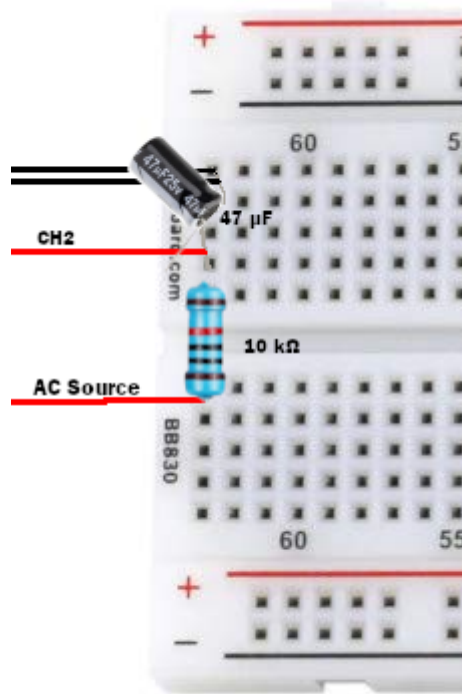


Figure 3.1.2: Breadboard setup used in this experiment.

3.2 Methods & Procedures

Our primary objective is to observe a charging/discharging pattern for the capacitor and fit it to a theoretical model (namely 2.0.5 for a charging fit and 2.0.10 for a discharging one).

In our specific case, an RC circuit was constructed using a $10\text{ k}\Omega$ resistor and a $47\text{ }\mu\text{F}$ capacitor in series. These values were validated using our DMM and were shown to correspond to the theoretical value. The voltage across the capacitor was monitored using an oscilloscope. Because the signal frequency was low, the oscilloscope was operated in roll mode. To isolate the transient behavior for measurement, the oscilloscope's *Run/Stop* function was utilized to freeze the display. We specifically chose to capture and analyze a discharging curve, as we observed it consistently yielded the cleanest signal.

Data collection was performed manually using the oscilloscope's built-in cursor

functionality. Approximately 30 discrete points were sampled along the discharge curve, recording both the time coordinate and the corresponding voltage across the capacitor. The measurement uncertainties were dictated by the instrument’s digital limits at our chosen zoom scale: the horizontal (time) precision was bounded at 0.008 s (the minimum possible cursor translation), and the vertical (voltage) precision was 0.01 V (limited by the two-digit digital readout).

During data analysis, the time coordinates were linearly shifted such that $t = 0$ aligned with the start of the discharge (the maximum observed voltage on the capacitor). Finally, the experimental RC time constant, τ , was determined by fitting the shifted time-voltage data to an exponential decay model using the `scipy.optimize` module in Python. Since we elected to analyze a discharge, we fit to 2.0.10.

From this we primarily hope to isolate the time constant τ . As defined in 2.0.4, τ is expected to be RC ; plugging in $R = 10 \pm 0.1 \text{ k}\Omega$ and $C = 47 \pm 2.35 \text{ }\mu\text{F}$ as outlined in the setup, we find that we expect $\tau = 470 \pm 24 \text{ ms} = 0.470 \pm 0.024 \text{ s}$. We further expect that $A = 1 \pm 0.001$ (with nominal uncertainty as prescribed).

3.3 Analysis

The normalized amplitude V_C/V_0 was fit to 2.0.10 using `scipy.optimize.curve_fit` over our datapoints, with time (shifted such that the first point was at $t = 0$) as the independent variable. Conducting OLS with Python, we found the best fit parameters to be:

$$A = 0.984 \pm 0.006, \quad (3.3.1)$$

$$\tau = 0.613 \pm 0.006 \text{ s} \quad (3.3.2)$$

Using our agreement tests in 2.0.26, we find that neither value agrees with its theoretical value of 1 and 0.470 s respectively. Noting that the reduced $\tilde{\chi}^2$ of the regression was 1.28, we conclude that this error is not emergent from a poor fit or failed modeling. To accommodate these errors, we analyze the regression graphs and residual plots.

The amplitude plot, depicting V_C/V_0 against time, with the best-fit curve is shown in 3.3.1. The same data is plotted on a semilog scale in 3.3.2.

Visually, the fit seems strong at all times, confirmed by having high R^2 and low $\tilde{\chi}^2$. Analyzing the residual plot in 3.3.1, we notice that the regression values consistently overpredict observations for $0 < t < 1 \text{ s}$ while underpredicting past that, despite the seemingly strong fit. This issue is more visible on the semilog scale in 3.3.2.

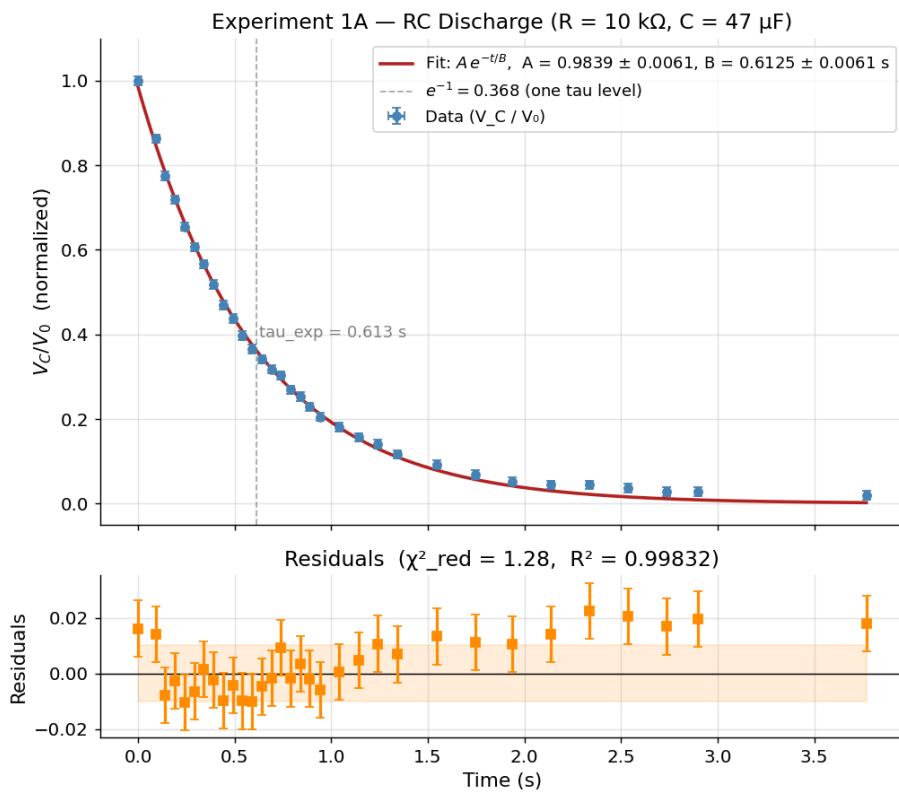


Figure 3.3.1: Graph of V_C/V_0 against time with least squares regression & residuals.

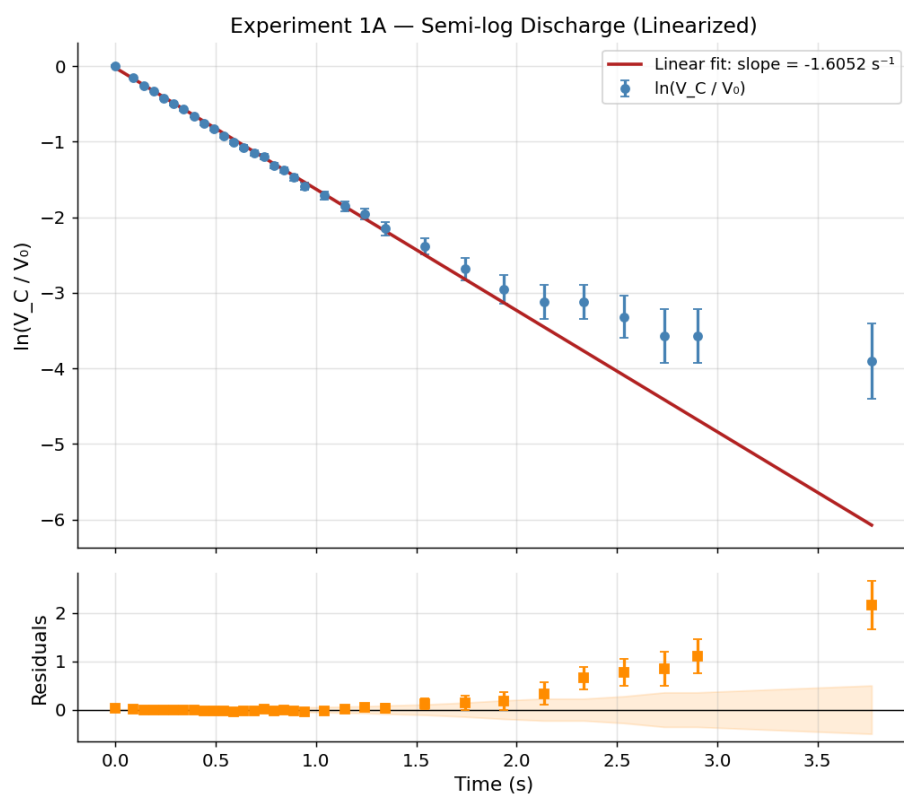


Figure 3.3.2: Graph of $\ln(V_C/V_0)$ against time with least squares regression & residuals.

3.4 Discussion

3.4.1 Addressing Regression Parameters & Disagreement

Neither the observed value of A nor B matched the predicted value of $A = 1$ or $\tau = 0.470$ s.

For A , there are several possibilities for why it underestimated the expected value; it seems that the capacitor only charges to about 98% of the AC source's voltage. The first and most likely explanation deals with a minor modeling error: the loading effect of the oscilloscope. The oscilloscope's $1\text{ M}\Omega$ internal input impedance acts in parallel with the capacitor, forming a voltage divider with the $10\text{ k}\Omega$ series resistor and the $50\text{ }\Omega$ internal resistance of the function generator. The maximum voltage that can be achieved by this will be:

$$V_{max} = V_0 \left(\frac{1\text{ M}\Omega}{1\text{ M}\Omega + 10\text{ k}\Omega + 50\Omega} \right) \approx 0.99V_0$$

which is more consistent with our observations. Other explanations include general statements about parasitic capacitance and inductance in the breadboard resulting in lower-than-expected voltages. An additional proposed explanation was that not enough time passed for the capacitor to charge fully. However, since the wave was set to 100 mHz , we had one cycle every 10 seconds; since τ was found to be about 0.613, this would give us approximately 8.3τ for a charge/discharge process, which would result in achieving $1 - e^{-8.3} \approx .9997$ of the full charge, so this explanation is unlikely.

3.4.2 Fit & Residual Failures

We generally observe a low reduced chi-square value and a high R^2 value, implying that the fit is quite strong. From an analysis of the residual plots, especially in 3.3.2, it is clear that the model overpredicts for small values of t and underpredicts for larger values of t , resulting in a non-random residual plot.

This may also be due to a modeling error: note that the model we use, $V = V_0(1 - e^{-t/\tau})$, assumes that the voltage across the capacitor will settle at 0 after infinite time. However, the oscilloscope may have a baseline offset caused by miscalibration, ambient noise, or incomplete discharging (potentially non-exponential discharging as time goes on). If we reconstruct the model under this hypothesis, we get a much more random residual plot as depicted in 3.4.1 and 3.4.2, and indeed a significantly lower reduced chi-squared and higher R^2 .

Taking the model to be $V_C/V_0 = A(1 - e^{-t/\tau}) + C$, we get that $C = 0.023 \pm 0.004\text{ V}$, which is small but not in agreement with 0. Note however that this model predicts $A = 0.972 \pm 0.007$ and $\tau = 0.574 \pm 0.009\text{ s}$, which are still not in

agreement with their theoretical values; explanations for those still correspond to those given in the previous section.

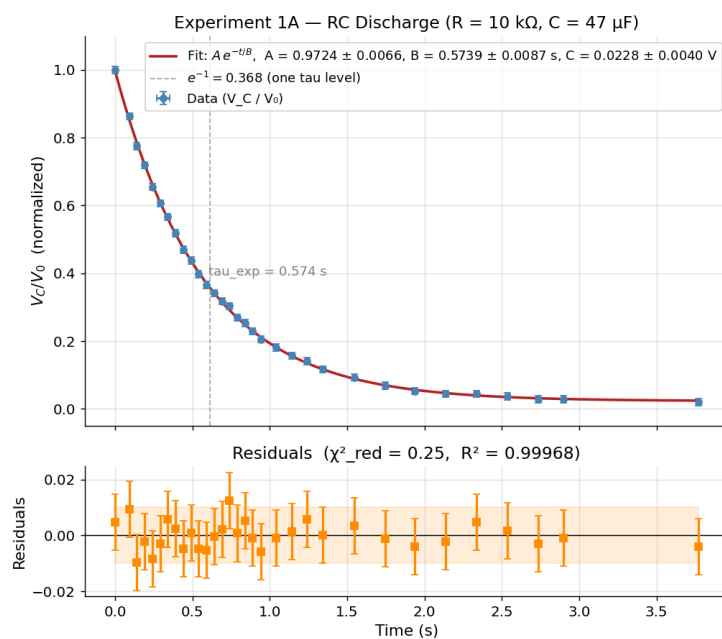


Figure 3.4.1: Regression & residual plot of V_C/V_0 , using our modified model hypothesis.

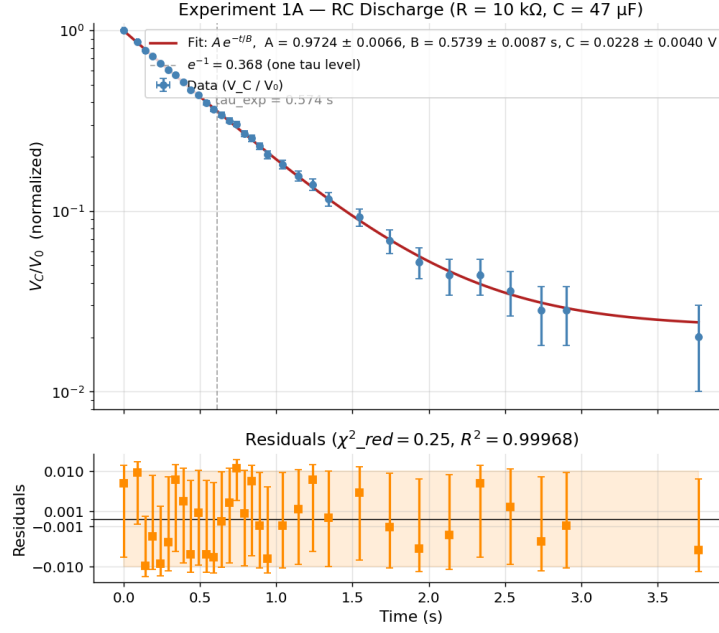


Figure 3.4.2: Regression & residual plot of V_C/V_0 on a semilog scale, using our modified model hypothesis.

3.5 Summary

An experiment was conducted to determine the time constant, τ , of an RC circuit comprising a $10 \text{ k}\Omega$ series resistor and a $47 \text{ }\mu\text{F}$ capacitor. A 100 mHz , 10 V_{pp} square wave with a $+5 \text{ V}$ offset was supplied via a function generator, and the capacitor's transient voltage was monitored using an oscilloscope in roll mode. Approximately 30 discrete data points were manually sampled along the discharge curve. Following a time-shift to align $t = 0$ with the maximum voltage, the normalized amplitude data (V_C/V_0) was fitted to an exponential decay model utilizing Ordinary Least Squares (OLS) regression via Python's `scipy.optimize`.

Theoretical calculations predicted a normalized initial amplitude of $A = 1$ and a time constant of $\tau = 0.470 \pm 0.024 \text{ s}$. However, the initial regression yielded $A = 0.984 \pm 0.006$ and $\tau = 0.613 \pm 0.006 \text{ s}$. Despite a relatively low reduced chi-square ($\tilde{\chi}^2 = 1.28$), neither parameter agreed with theoretical expectations.

The discrepancy in amplitude ($A < 1$) is primarily attributed to the loading effect of the oscilloscope; its $1 \text{ M}\Omega$ internal input impedance acts in parallel with the capacitor, creating a voltage divider that theoretically caps the maximum

voltage at $\approx 0.99V_0$. Furthermore, residual analysis revealed a non-random distribution, with the model overpredicting early times ($t < 1$ s) and underpredicting later times. This suggested a systemic baseline offset. Introducing a constant offset parameter ($C = 0.023 \pm 0.004$ V) to the model resolved the residual patterns and adjusted the fit parameters to $A = 0.972 \pm 0.007$ and $\tau = 0.574 \pm 0.009$ s. While this improved the model's validity, the parameters remained outside strictly predicted bounds, likely due to unmodeled parasitic circuit characteristics.

4 Experiment 1B: Handcrafting a Capacitor

4.1 Setup

Our objective in this portion of the experiment is to construct a homemade capacitor and characterize its behavior by determining its capacitance through RC circuit analysis.

4.1.1 Materials

For the construction and testing of the homemade capacitor, we utilized:

- Aluminum foil,
- Standard letter paper (acting as the dielectric),
- Glue,
- Jumper wires and alligator clips,
- A digital multimeter (DMM),
- A 1 M Ω resistor,
- An oscilloscope and function generator (as used in Experiment 1A).

4.1.2 Capacitor Construction

The homemade capacitor was constructed as a parallel-plate capacitor. Two pieces of aluminum foil were cut and carefully glued to opposite sides of a single sheet of printer paper. Special care was taken to ensure the foils were flat and adhered securely to the paper. We also ensured that no pinholes or connections between sides were present in the paper, as this would short-circuit the device.

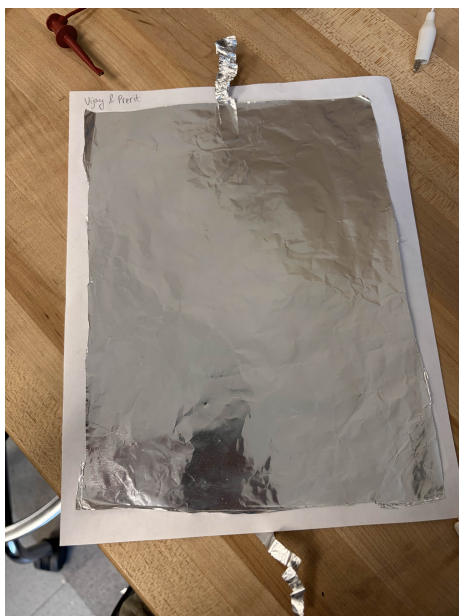


Figure 4.1.1: The homemade capacitor we constructed.

Electrical connections were established by taping an additional small stretch of aluminum foil to each end and clipping onto them with alligator clips. We initially made a glue-based connection but then realized that the glue would insulate the connection, so we switched to a tape-based approach (not pictured). An initial capacitance reading was also taken with the DMM to serve as a baseline, giving 0 nF, implying our capacitance was too small to measure accurately.

4.1.3 Circuit Construction & Measurement

To determine the capacitance experimentally, an RC circuit was constructed using the homemade capacitor and the chosen resistor ($1\text{ M}\Omega$). We chose to use a larger resistor to minimize the impact of the internal resistances in our homemade capacitor and result in a cleaner capture overall. Similar to the procedure in Experiment 1A, the circuit was subjected to an AC voltage source, done at 5 Vpp with a 5V offset. The voltage across the homemade capacitor was monitored using an oscilloscope to capture the charging and discharging curves. By capturing these curve points, we aimed to obtain the time constant τ to fully characterize the behavior of our handcrafted component.

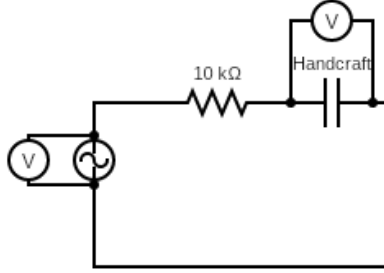


Figure 4.1.2: Circuit diagram for Experiment 1B.

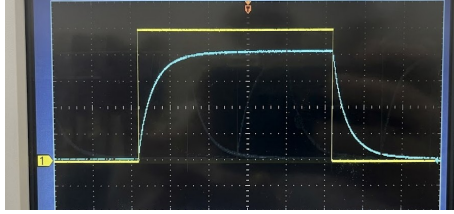


Figure 4.2.1: An image of the reading we used for our analysis.

4.2 Methods & Procedures

Before taking empirical measurements, an estimation for the capacitance of our homemade capacitor was calculated. Modeling the device as an ideal parallel-plate capacitor, the theoretical capacitance is given by 2.0.12. We took our paper to be 21 cm $\times$ 30 cm, and gave approximately 1 centimeter of "breathing room" on each end to assure that the aluminum foil sheets did not touch, giving dimension of 19 ± 0.1 cm $\times$ 28 ± 0.1 cm, assuming precision errors to the accuracy of our ruler. Assuming the dielectric constant of standard printer paper is approximately $\epsilon_r \approx 3.8 \pm 0.1$ based on online estimates, and using our measured foil area of $A = 532 \pm 3$ cm² and an estimated paper thickness of $d = 10^{-4} \pm 10^{-5}$ m, our theoretical prediction was $C_{\text{est}} = 18 \pm 1.8$ nF.

Following construction, we waited 2 days before taking measurements to give the glue time to dry. We then took an initial baseline measurement of the capacitance using the digital multimeter (DMM), which yielded a value of 11.3 nF; possible sources for this variation will be discussed later.

To empirically determine the actual capacitance, we integrated the homemade capacitor into the RC circuit discussed in the setup, using a $R = 1$ M Ω resistor. We recorded the charging and discharging curves using the oscilloscope. As

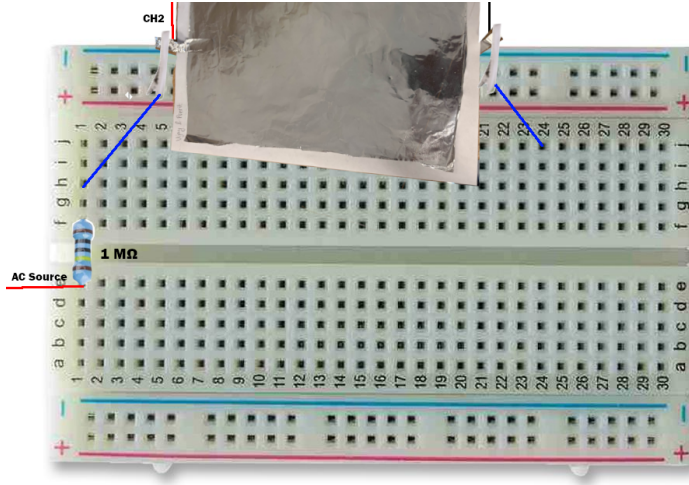


Figure 4.2.2: Crude depiction of our breadboard setup in Experiment 1B.

exponential behavior is defined by the time constant τ , obtaining five reliable data points along the curve was sufficient to characterize the transient behavior and perform an exponential regression.

4.3 Analysis

Our data was fit to 2.0.5 using `scipy.optimize.curve_fit` over our datapoints, with time (shifted such that the first point was at $t = 0$) as the independent variable. Conducting OLS with Python, we found the best fit value of τ to be

$$\tau = 0.32 \pm 0.02 \text{ s}$$

Given our known resistance of $1 \text{ M}\Omega \pm 1\%$, we can calculate our experimental value of capacitance to be

$$C_{\text{exp}} = \frac{\tau_{\text{exp}}}{R} = 32 \pm 2 \text{ nF}$$

which does not agree with our theoretical value of $18 \pm 1.8 \text{ nF}$.

The reached capacitor voltage is depicted in 4.3.1. Note the low value of $\tilde{\chi}^2$, the high value of R^2 , and the relative random scatter in the residual plot imply that 2.0.10 is an accurate description of the phenomenon we are measuring, though the capacitor does not seem to charge fully, with the maximum voltage not quite reaching the 5 V expected. Also note that while the scatter in the

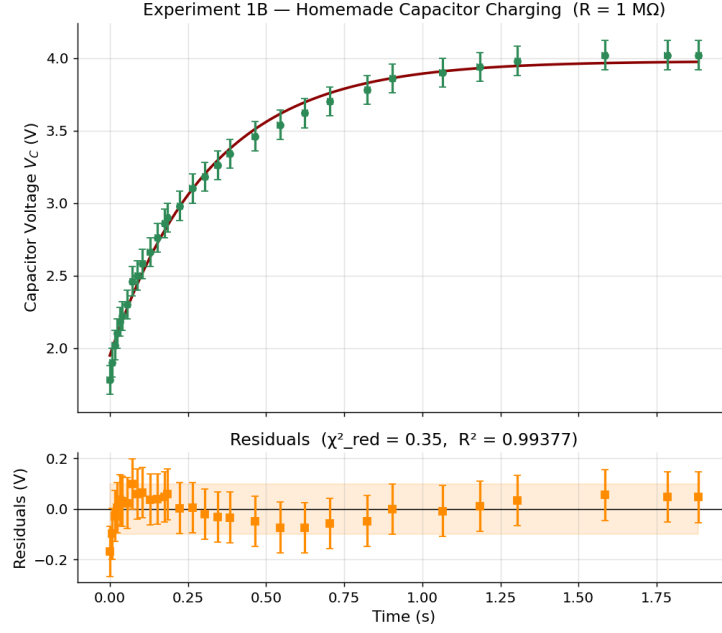


Figure 4.3.1: V_C plotted against time, regressed to fit 2.0.10.

residuals initially seems mostly random, there seems to be an almost oscillatory pattern emerging; most accurately, the model's error at time t seems to be correlated with the errors at times around t .

Using our empirically derived capacitance C_{exp} and the known overlapping area of our aluminum foil plates A , we can rearrange the parallel-plate capacitor equation to estimate the *effective* thickness of our paper dielectric, d_{eff} :

$$d_{\text{eff}} = \frac{\epsilon_r \epsilon_0 A}{C_{\text{exp}}} = 0.06 \pm 0.01 \text{ mm} \quad (4.3.1)$$

which is smaller than the true separation of 0.1 mm.

4.4 Discussion

4.4.1 High Observed Capacitance

Our experimental capacitance of 32 nF does not agree with our theoretical capacitance, being approximately 1.6 times our predicted value. The most likely culprit for this is a layer of glue between the foil and the paper having a significantly higher dielectric constant than paper's. This would result in the true

value of ϵ_r in 2.0.12 being significantly higher and thereby result in a higher capacitance.

4.4.2 Effective Dielectric Thickness

This was found to be slightly more than half of the true separation, aligning with our experimental capacitance being higher than our theoretical capacitance. It is unlikely that this reflects something physical about the separation as much as it reflects an incorrect estimation of ϵ_r .

4.4.3 Oscillatory Residuals

There are numerous possible explanations for this, including simple random error. Other similar errors in this vein would emerge from machine error, with the oscilloscope generally overreporting and underreporting in neighborhoods instead of randomly. There are also physical explanations. The capacitance of our device may not have been constant throughout our analysis, or there may have been secondary effects emerging from soakage into the paper. Among these, I believe the most likely explanation is some flaw in the capacitor's construction leading to high soakage, as the effect is significant enough to warrant a physical explanation and that is the most likely among them.

4.4.4 Industrial Methods for High Capacitance

Our handcrafted capacitor demonstrates that achieving a large capacitance with flat sheets of foil and paper is highly inefficient in terms of volume. To pack substantial capacitances (e.g., 1 mF to 1 F) into small commercial packages, the electronics industry relies on two primary geometric manipulations: maximizing surface area (A) and minimizing dielectric thickness (d).

For standard electrolytic capacitors (often in the millifarad range), manufacturers use etched aluminum foil, which drastically increases the microscopic surface area. Instead of paper, the dielectric is an infinitesimally thin layer of aluminum oxide (Al_2O_3) created via an electrochemical anodization process. These thin, high-area layers are then tightly rolled into a compact cylinder. For supercapacitors (achieving 1 F or more), the design leverages the electric double-layer effect. They utilize highly porous materials like activated carbon, which provide an astronomically large internal surface area, paired with an effective charge-separation distance (dielectric thickness) on the scale of nanometers, allowing for massive energy storage in a diminutive footprint.

4.5 Summary

To characterize our homemade parallel-plate capacitor, we constructed the device by gluing aluminum foil sheets to opposite sides of a standard sheet of printer paper (acting as the dielectric) and integrated it into an RC circuit with a $1\text{ M}\Omega$ resistor driven by a 5 V_{pp} square wave with a 5 V offset. Using `scipy.optimize.curve_fit` to perform an exponential regression on the oscilloscope-captured charging/discharging curves, we extracted a time constant of $\tau = 0.32 \pm 0.02\text{ s}$, yielding an experimental capacitance of $C_{\text{exp}} = \tau/R = 32 \pm 2\text{ nF}$. This exceeds both the DMM baseline reading of 11.3 nF and the theoretical estimate of $C_{\text{est}} = 18 \pm 1.8\text{ nF}$, computed from the parallel-plate relation $C = \epsilon_r \epsilon_0 A/d$ using $\epsilon_r \approx 3.8$, $A = 532 \pm 3\text{ cm}^2$, and $d = 0.1 \pm 0.01\text{ mm}$.

The discrepancy between C_{exp} and C_{est} is attributed primarily to the glue layer between the foil and paper, which likely has a higher dielectric constant than paper alone, effectively increasing ϵ_r and thus the capacitance. Rearranging the parallel-plate equation with C_{exp} yields an effective dielectric thickness of $d_{\text{eff}} = 0.06 \pm 0.01\text{ mm}$, roughly half the true paper thickness, consistent with this interpretation. The regression achieved a high R^2 and low $\tilde{\chi}^2$, though residuals exhibited a mild oscillatory pattern suggestive of dielectric soakage into the paper, wherein charge slowly migrates into the bulk dielectric and distorts the ideal exponential response.

5 Experiment 2A: RLC Underdamped Transient Response

5.1 Setup

Our objective in this experiment is to showcase the RLC underdamped transient response.

5.1.1 Materials

For this section of the lab, we used:

- an MSO 2024B Oscilloscope,
- a Tektronix AFG2021 Function Generator,
- a breadboard,
- two $100\ \Omega$ resistors with a $\pm 1\%$ tolerance (wired in parallel to achieve $50\ \Omega$),
- a $0.1\ \mu\text{F}$ capacitor with a $\pm 5\%$ tolerance,
- a $100\ \text{mH}$ inductor with a $\pm 10\%$ tolerance,
- and various leads, grabbers, and BNC cables.

5.1.2 Function Generator

The objective was to create a low-frequency square wave to serve as our power source. To do so, we:

- chose “Square Wave” as our desired signal output,
- set the frequency to $10\ \text{Hz}$,
- set the output load impedance to Hi-Z,
- set the amplitude to $4\ \text{V}_{\text{pp}}$,
- and connected the function generator output directly to the circuit and to CH1 of the oscilloscope.

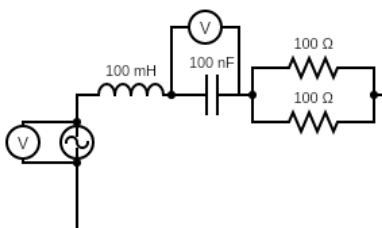


Figure 5.1.1: Circuit diagram for Experiment 2A.

5.1.3 Oscilloscope

Here, we hoped to measure both the function generator output and the voltage difference across the resistors. To accomplish this, we:

- displayed channels 1 and 2 on the scope,
- connected one BNC cable from the function generator output to the oscilloscope CH1 input,
- connected a BNC cable from CH2 to a mini-grabber across the resistors,
- and set the oscilloscope to Auto trigger mode. Since the signal frequency was 10 Hz, the oscilloscope could trigger repetitively in Auto mode, unlike in Experiment 1A. We used the Run/Stop button to freeze the display on a rising or falling edge of interest.

5.1.4 RLC Circuit Construction

We constructed the series RLC circuit depicted in 5.1.1 on the breadboard using a 100 mH inductor, a $0.1\ \mu\text{F}$ capacitor, and a $50\ \Omega$ effective resistance (achieved by wiring two $100\ \Omega$ resistors in parallel). Before assembling the circuit, we measured the internal resistance of the inductor using the DMM and recorded $R_L = 71.8\ \Omega$, which was included in the total circuit resistance for all subsequent calculations. CH1 monitored the function generator output and CH2 monitored the voltage across the resistors, allowing us to observe the damped “ringing” transient behavior at both the rising and falling edges of the square wave.

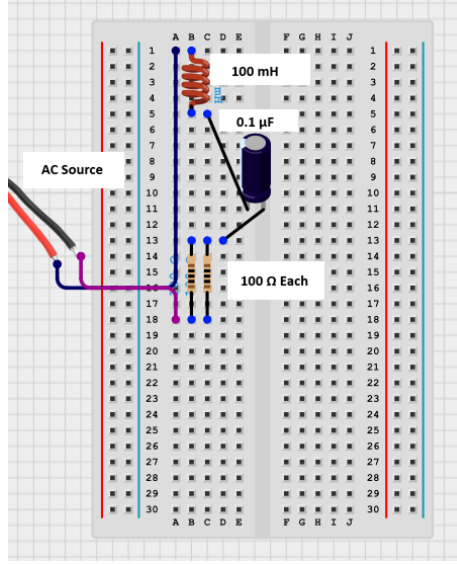


Figure 5.2.1: Breadboard diagram for Experiment 2A.

5.2 Methods & Procedures

In Experiment 2A, our objective was to explore the underdamped transient response of an RLC circuit and extract the characteristic parameters α and ω_0 by fitting the observed resistor voltage to the underdamped current model in 2.0.18.

We first verified that the circuit was in the underdamped regime by computing the theoretical values of ω_0 and α from 2.0.16. Using $L = 100 \text{ mH}$, $C = 0.1 \mu\text{F}$, and $R_{\text{total}} = R + R_L = 50 + 71.8 = 121.8 \Omega$, we obtain theoretical values of $\omega_0 = 10000 \pm 600 \text{ rad/s}$ and $\alpha = 610 \pm 25 \text{ rad/s}$, confirming $\alpha \ll \omega_0$ and hence underdamped behavior.

We observed damped “ringing” in the CH2 resistor voltage at the rising edge of the CH1 input square wave. To isolate the transient behavior for measurement, the oscilloscope’s Run/Stop function was utilized to freeze the display. We arbitrarily chose to capture and analyze a rising edge curve, as we observed no significant difference between the noise characteristics of a rising versus a falling edge signal.

Data collection was performed manually using the oscilloscope’s built-in cursor functionality, recording the peak voltages of the damped oscillation and the time of each successive peak. The measurement uncertainties were dictated by the instrument’s digital limits at our chosen zoom scale: the horizontal (time)

precision was bounded at 0.04 ms (the minimum possible cursor translation), and the vertical (voltage) precision was 0.01 V (limited by the two-digit digital readout). A total of 10 peaks were extracted. The baseline voltage V_{base} was subtracted from each peak voltage to isolate the oscillation amplitude.

During data analysis, the time coordinates were linearly shifted such that $t = 0$ aligned with the start of the transient pulse. The oscillation frequency ω was extracted from the average peak-to-peak spacing. We then fit the peak envelope $A_{\text{peak}}(t) = \tilde{A} e^{-\alpha t}$ to the extracted amplitudes using `scipy.optimize.curve_fit` to determine α_{exp} and $\tilde{A}$, and separately recovered ω_0 via $\omega_0 = \sqrt{\omega^2 + \alpha^2}$.

5.3 Analysis

During data analysis, the time coordinates were linearly shifted such that $t = 0$ aligned with the start of the transient pulse. The oscillation frequency ω was extracted from the average peak-to-peak spacing. We then fit the peak envelope $A_{\text{peak}}(t) = \tilde{A} e^{-\alpha t}$ to the extracted amplitudes using `scipy.optimize.curve_fit` to determine α_{exp} and $\tilde{A}$, and separately recovered ω_0 via $\omega_0 = \sqrt{\omega^2 + \alpha^2}$.

Conducting OLS with Python, we found the best-fit parameters to be:

$$\begin{aligned}\tilde{A} &= 3.96 \pm 0.07 \text{ V} \\ \alpha &= 700 \pm 18 \text{ rad/s} \\ \omega &= 10030 \pm 44 \text{ rad/s} \\ \omega_0 &= 10060 \pm 43 \text{ rad/s}\end{aligned}$$

Applying the agreement test in 2.0.26, all four fitted values agree with their respective theoretical predictions within two standard deviations. The decay constant α shows the largest relative deviation from its theoretical value, which is most visible on the semilog plot in 6.3.2. Inspection of the residual plots in both 6.3.1 and 6.3.2 reveals no systematic trend across the full data range; however, there is a slight systematic residual for $t > 4$ ms, consistent with the growing relative uncertainty at low amplitudes discussed below.

5.4 Discussion

5.4.1 Quality of the Fit

The decay envelope fit and the semilog representation of the peak amplitudes both appear visually strong. However, in the semilog plot, the low-amplitude peaks at late times exhibit disproportionately large error bars compared to the

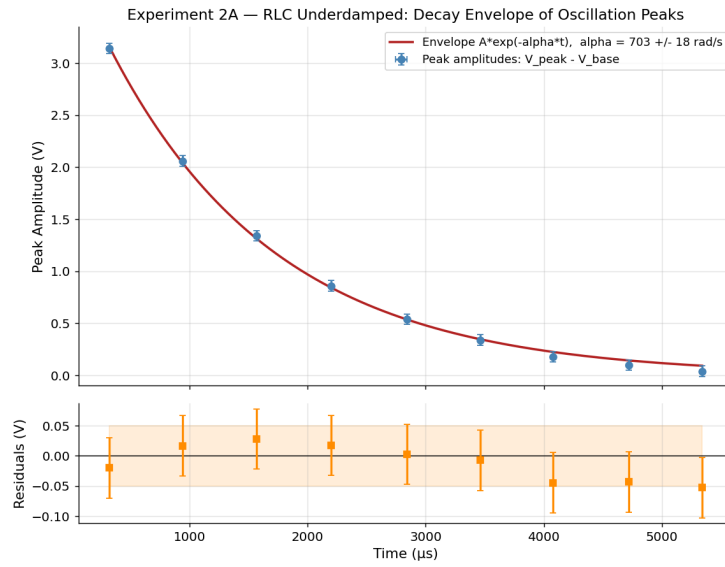


Figure 5.3.1: Peak amplitudes plotted against time, regressed to fit $A_{\text{peak}}(t) = \tilde{A} e^{-\alpha t}$ and residuals

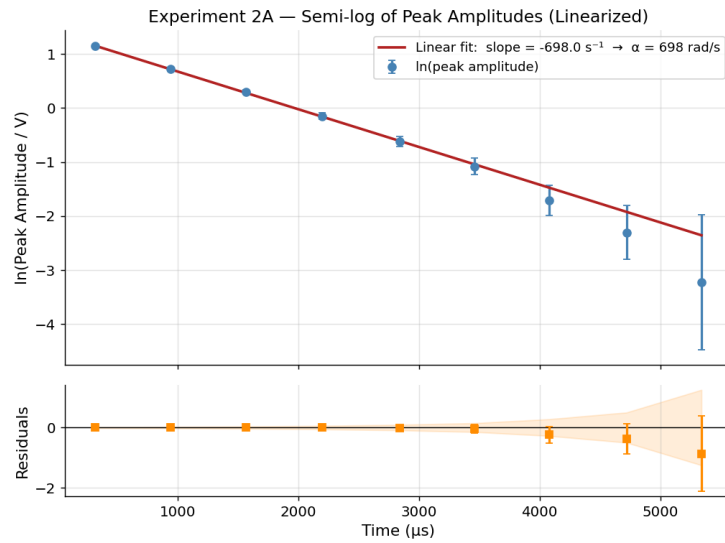


Figure 5.3.2: Graph of $\tilde{A}$ against time with least squares regression & residuals.

early peaks. This is because the fixed voltage uncertainty blows up after taking a log for small values.

5.4.2 Sources of Experimental Error

Several experimental factors are likely responsible for the residual scatter and the elevated deviation in α . First, any inaccuracy in measuring V_{base} (the offset voltage) is more significant when V is very small, causing the baseline-subtracted peaks to be systematically mis-estimated. Second, the oscilloscope’s vertical resolution of 0.01 V was adequate for resolving the earliest, largest peaks but insufficient for the smallest late-time peaks. Additional contributions likely arose from manual cursor-based peak selection and non-ideal inductor losses not captured by a simple constant series resistance model.

5.5 Summary

To characterize the underdamped transient response of the series RLC circuit, we recorded the peak voltages and times of ten successive oscillation peaks from the CH2 resistor voltage at a rising edge of the 10 Hz square-wave input. After shifting the time axis to align $t = 0$ with the onset of the transient, we fit the exponential decay envelope $\tilde{A}e^{-\alpha t}$ to the extracted peak amplitudes using `scipy.optimize.curve_fit`, and independently determined ω from the mean peak-to-peak spacing, recovering $\omega_0 = \sqrt{\omega^2 + \alpha^2}$.

The best-fit parameters $\tilde{A} = 3.96 \pm 0.07$ V, $\alpha = 700 \pm 18$ rad/s, $\omega = 10030 \pm 44$ rad/s, and $\omega_0 = 10060 \pm 44$ rad/s all agree with theoretical predictions within two standard deviations under the agreement test in 2.0.26. The decay constant α deviates most from its theoretical counterpart, a result attributed to the non-ideal frequency-dependent losses in the inductor, manual cursor-based peak selection, limited oscilloscope vertical resolution (0.01 V), and sensitivity of the baseline subtraction to small errors in V_{base} at late times. Residuals were mostly random, though a small systematic error appeared for $t > 4$ ms where the signal-to-noise ratio was lowest.

6 Experiment 2B: RLC Overdamped Transient Response

6.1 Setup

Our objective in this experiment is to showcase the RLC overdamped transient response.

6.1.1 Materials

For this section of the lab, we used:

- an MSO 2024B Oscilloscope,
- a Tektronix AFG2021 Function Generator,
- a breadboard,
- a $1\text{ k}\Omega$ resistor with a $\pm 1\%$ tolerance,
- a $47\text{ }\mu\text{F}$ capacitor with a $\pm 5\%$ tolerance,
- a 100 mH inductor with a $\pm 10\%$ tolerance (with measured internal resistance $R_L = 71.8\text{ }\Omega$),
- and various leads, grabbers, and BNC cables.

6.1.2 Function Generator

The objective was to create a low-frequency square wave to serve as our power source. To do so, we:

- chose “Square Wave” as our desired signal output,
- set the frequency to 100 mHz ,
- set the output load impedance to Hi-Z,
- set the amplitude to 4 V_{pp} ,
- and connected the function generator output directly to the circuit and to CH1 of the oscilloscope.

The lower frequency (relative to Experiment 2A) was chosen to ensure each half-period was long enough for the overdamped transient to fully decay before the next voltage step.

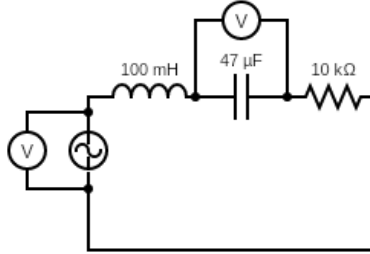


Figure 6.1.1: Circuit diagram for Experiment 2B.

6.1.3 Oscilloscope

Here, we hoped to measure both the function generator output and the voltage difference across the resistors. To accomplish this, we:

- displayed channels 1 and 2 on the scope,
- connected one BNC cable from the function generator output to the oscilloscope CH1 input,
- connected a BNC cable from CH2 to a mini-grabber placed across the resistors,
- and set the oscilloscope to Auto trigger mode.

6.1.4 RLC Circuit Construction

We constructed the series RLC circuit in 6.1.1 on the breadboard using the same 100 mH inductor as in Experiment 2A (with $R_L = 71.8\ \Omega$), a $47\ \mu\text{F}$ capacitor, and a $1\ \text{k}\Omega$ resistor. The total circuit resistance was therefore $R_{\text{total}} = 1000 + 71.8 = 1071.8\ \Omega$. CH1 monitored the function generator output and CH2 monitored the voltage across the resistors. The overdamped regime was confirmed a priori by computing α and ω_0 from 2.0.16: with these component values $\alpha \gg \omega_0$, so no oscillatory behavior was expected.

6.2 Methods & Procedures

In Experiment 2B, our objective was to explore the overdamped transient response of an RLC circuit and extract the effective decay constant by fitting the

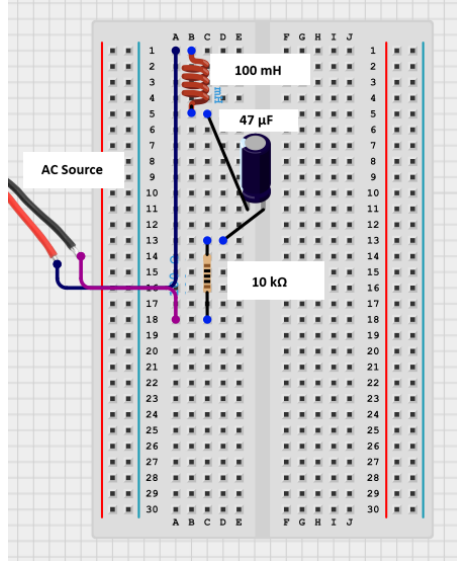


Figure 6.2.1: Breadboard diagram for Experiment 2B.

observed resistor voltage to an exponential decay model.

We first verified that the circuit was in the overdamped regime by computing the theoretical parameters from 2.0.16. Using $L = 100 \text{ mH}$, $C = 47 \mu\text{F}$, and $R_{\text{total}} = 1071.8 \Omega$, we find $\omega_0 = 1/\sqrt{LC}$ and $\alpha = R_{\text{total}}/(2L)$, confirming $\alpha > \omega_0$ and hence overdamped behavior.

We observed an aperiodic exponential decay in the CH2 resistor voltage at both rising and falling edges of the CH1 input square wave, consistent with the overdamped regime. To isolate the transient for measurement, the oscilloscope's Run/Stop function was used to freeze the display.

Data collection was performed manually using the oscilloscope's built-in cursor functionality. The voltage across the resistors was recorded at multiple time points along the decay curve, providing enough points to characterize the decay constant. As noted in the lab manual, the $t = 0$ start time required careful identification, since the transient onset is not always sharply defined in the overdamped case. We set $t = 0$ at the point of the voltage step transition on CH1 and measured subsequent CH2 values relative to that reference.

During data analysis, the time coordinates were shifted such that $t = 0$ aligned with the start of the transient. The normalized resistor voltage $V_R(t)/V_0$ was fit to a single exponential decay model,

$$\frac{V_R(t)}{V_0} = A e^{-Dt}, \quad (6.2.1)$$

using `scipy.optimize.curve_fit`, to extract the fit amplitude A and fit decay constant D , along with their associated uncertainties. The fitted decay constant D was then compared to the theoretical slow-mode decay rate $D_{\text{slow}} = \alpha - b$ using the agreement test in 2.0.26, where $b = \sqrt{\alpha^2 - \omega_0^2}$.

6.3 Analysis

During data analysis, the time coordinates were shifted such that $t = 0$ aligned with the start of the transient. The normalized resistor voltage $V_R(t)/V_0$ was fit to the single exponential decay model in 6.2.1 using `scipy.optimize.curve_fit`, extracting the fit amplitude A and fit decay constant D along with their associated uncertainties.

We perform an agreement test of D against the theoretical slow-mode decay rate $D_{\text{slow}} = \alpha - b$ (where $b = \sqrt{\alpha^2 - \omega_0^2}$) rather than D_{fast} , because the fast mode decays on a timescale of $\tau_{\text{fast}} = 1/D_{\text{fast}} \approx 0.09$ ms, so it is entirely negligible after $5\tau_{\text{fast}} \approx 0.45$ ms. Since our data spans 226 ms, a single exponential with decay constant D_{slow} captures the dominant behavior throughout the measurement window.

Conducting OLS with Python, we found the best-fit parameters to be:

$$\begin{aligned} A &= 1.028 \pm 0.005 \\ D &= 15.5 \pm 0.2 \text{ rad/s} \\ \tau &= 64.674 \text{ ms} \end{aligned}$$

It is important to note that agreement between D_{exp} and D_{slow} is partly trivial, as the theoretical uncertainty on D_{slow} is very large owing to the near-cancellation of α and b when computing $\alpha - b$. The reduced chi-squared value of $\tilde{\chi}^2 = 2.15$ is likely higher than 1 due to the effects of D_{fast} dominating at the start of the regime. To further accommodate these errors, we analyze the regression graphs and residual plots.

Visually, the fit appears strong across the full time range, consistent with the high R^2 and relatively low $\tilde{\chi}^2$. Analyzing the residual plot in 6.3.1, we notice that the regression values consistently overpredict observations for $2.5 \leq t \leq 10$, while underpredicting past that. Moreover, this issue is more visible on the semilog scale in 6.3.2.

Note that the relatively low value of $\tilde{\chi}^2$, the high value of R^2 , and the relative random scatter in the residual plot imply that 6.3.1 is an accurate description of the phenomenon we are measuring. However, while the scatter in the residuals initially seems mostly random, there seems to be an almost oscillatory pattern emerging, not unlike that in 4.3.1. Indeed, the regression values consistently overpredict observations for $2.5 < t < 10$, while underpredicting past that.

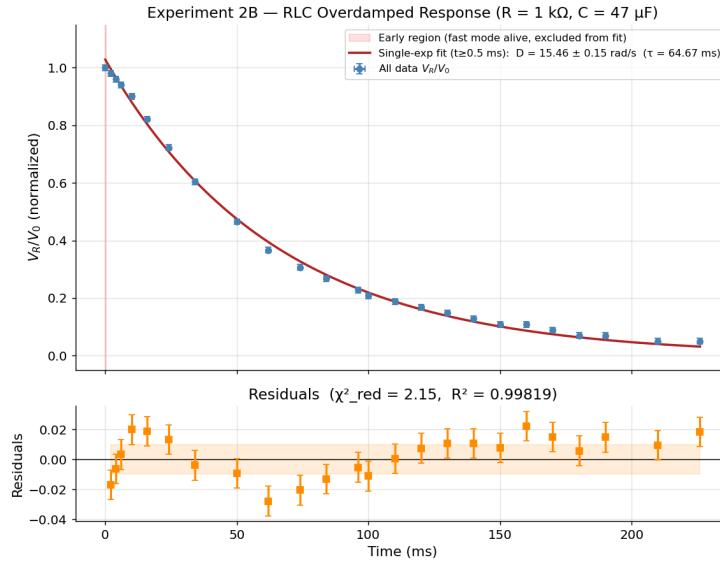


Figure 6.3.1: Normalized V_R/V_0 plotted against time, regressed to fit 6.2.1 and residuals.

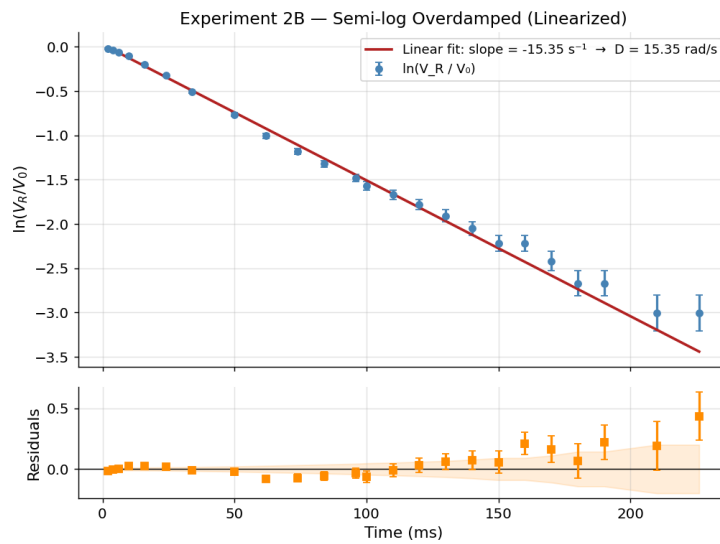


Figure 6.3.2: Graph of V_R/V_0 on a log scale against time with least squares regression & residuals.

6.4 Discussion

6.4.1 Quality of the Fit

The overdamped exponential fit and its semilog representation are visually strong, with a high R^2 and an acceptable $\tilde{\chi}^2$. Nevertheless, as with Experiment 2A, the lower-amplitude data points at late times deviate more prominently from the semilog fit due to the larger relative errors. In this case, deviations from linearity in the semilog plot arise not only from experimental uncertainty rising at low amplitudes due to oscilloscope resolution and noise (with log amplifying them), but also from the fact that the transient decay is meant to be a superposition of two exponential modes. The single-exponential model is only valid once the fast mode has decayed, which explains why the fit is most accurate at intermediate times.

6.4.2 Oscillatory Residuals

There are numerous possible explanations for this, including simple random error. Other similar errors in this vein would emerge from machine error, with the oscilloscope generally overreporting and underreporting in neighborhoods instead of randomly. Moreover, there could be residual inductor parasitic properties that introduce a small oscillatory correction to the purely overdamped response.

6.5 Summary

To characterize the overdamped transient response of the series RLC circuit, we recorded the resistor voltage $V_R(t)$ at multiple time points along the aperiodic decay following a step input, with $t = 0$ set at the rising edge of the 100 mHz square-wave input. The normalized voltage $V_R(t)/V_0$ was fit to the single-exponential model Ae^{-Dt} using `scipy.optimize.curve_fit`, yielding best-fit parameters $A = 1.028 \pm 0.0052$ and $D_{\text{exp}} = 15.5 \pm 0.150$ rad/s (corresponding to $\tau_{\text{exp}} = 64.674$ ms). The fitted decay constant was compared to the theoretical slow-mode rate $D_{\text{slow}} = \alpha - b$, and agreement was confirmed.

The fit quality was high overall (R^2 close to 1, $\tilde{\chi}^2 = 2.15$), though the slightly elevated $\tilde{\chi}^2$ and the systematic residual pattern — overprediction for $2.5 < t < 10$ ms and underprediction thereafter — are consistent with the fast exponential mode contributing residual curvature at early times that a single-exponential model cannot accommodate. In the semilog representation, late-time data points exhibit larger relative deviations, attributed to oscilloscope resolution limits amplified by the logarithmic scale.

7 Conclusion

In Experiment 1A, the discharge time constant of an RC circuit ($R = 10\text{ k}\Omega$, $C = 47\text{ }\mu\text{F}$) was measured as $\tau = 0.613 \pm 0.006\text{ s}$, which did not agree with the theoretical prediction of $0.470 \pm 0.024\text{ s}$. The shortfall in amplitude ($A = 0.984$) was attributed primarily to the oscilloscope’s loading effect, while the residual non-randomness was resolved by introducing a small baseline offset term, consistent with a minor oscilloscope miscalibration or incomplete discharge. A model considering an offset voltage was explored and was found to have generally lower error and saw a more random scatter in its residual plot. To develop this experiment further, we especially propose using more precise equipment to prevent the issues we described.

In Experiment 1B, a homemade parallel-plate capacitor was constructed from aluminum foil and printer paper and characterized via the same RC method. The measured capacitance of $C_{\text{exp}} = 32 \pm 2\text{ nF}$ exceeded the theoretical estimate of $C_{\text{est}} = 18 \pm 1.8\text{ nF}$, most likely due to the higher dielectric constant of the glue layer between the foil and paper. An effective dielectric thickness of $d_{\text{eff}} = 0.06 \pm 0.01\text{ mm}$ was calculated, approximately half the true paper thickness, consistent with this interpretation. To enhance the outcome of this experiment, we recommend an exploration of the dielectric constant of the dried glue or the paper-glue combination to more accurately gauge the emergent dielectric constant, in addition to using more precise equipment.

In Experiment 2A, the underdamped RLC transient was analyzed by fitting the exponential decay envelope of the oscillation peaks. All four extracted parameters ($\tilde{A}$, α , ω , and ω_0) agreed with their theoretical values within two standard deviations, though the decay constant α showed the largest relative deviation, attributed to non-ideal inductor losses, limited oscilloscope resolution, and manual peak selection. We then propose the usage of more precise, higher-resolution equipment, particularly at low voltages, and a more careful measurement of baseline voltage.

In Experiment 2B, the overdamped RLC transient was fit to a single exponential decay, yielding a slow-mode decay constant of $D_{\text{exp}} = 15.5 \pm 0.15\text{ rad/s}$, which agreed with the theoretical slow-mode rate within its (large) uncertainty. For this experiment we suggest more precise equipment to reduce those large uncertainties, especially higher resolution on low voltage readings. We could have also tried to use an exponential decay model considering D_{fast} as well as D_{slow} to improve modeling accuracy.

Across all experiments, a recurring theme was the amplification of small absolute errors under logarithmic transformation, producing disproportionately large apparent deviations at late times in semilog plots. Oscilloscope resolution limits and manual cursor-based data collection were consistent sources of uncertainty throughout.

8 References

[1] University of California Berkeley, Physics 5BL. *Tier 1 Lab: RC and RLC Circuits*; Berkeley, CA, 2026

We used artificial intelligence (specifically Claude) for code debugging and to proofread the report.

9 Author Contributions

Data collection split evenly amongst members.

- **Vijay Hans:** Experiment 1 Writing/Analysis/Theory, Introduction
- **Prerit Bucha:** Experiment 2 Writing/Analysis/Theory, Conclusion